



CS 1002 – Programming Fundamentals (Fall 2022) Assignment 2

Topics Covered: Problem solving, pseudocode and flowchart	Submission Deadline: <i>Friday – September 30, 2022 by 16.00 sharp</i>
Submission Guidelines: - Submit your own original work. Plagiarism will not be tolerated, either done from the internet or from any fellow. - Submit your hand-written solutions in your respective instructor's office.	

Write pseudocode and draw flowchart for each of the problem given below. You are not allowed to use logical operators (i.e., AND or OR) for any purpose.

Problem # 1:

A private electricity company needs an application that allows their customers to calculate their electricity bills. A customer will input units consumed and his/her bill will be calculated according to the following criteria:

- For first 50 units Rs. 8/unit
- For next 100 units Rs. 10/unit
- For next 100 units Rs. 14/unit
- For unit above 250 Rs. 18/unit

Furthermore, fuel surcharges of 20% and Govt. Tax of 10% will also be included to a customer's bill. Calculate and display the total bill of a customer.

Problem # 2:

Amjad goes to school on weekdays (Monday to Friday). He has got a small money box from which he gets his expenses for the round trip of the school daily. He puts his hand in the box and pulls some notes. If the total amount of note is above 40 rupees, he travels by rickshaw. If the amount is between 20-40, he travels by bus. Otherwise, he goes on bicycle. Rickshaw fare is 15 rupees and bus costs 8 rupees for one sided travel. In the school he eats candies for all the remaining amount that is left after the travelling expenses. Each candy costs 3 rupees. Print the amount spent on travelling and candies. If some amount is left behind, then it is put back in the money box.

Problem # 3:

An insurance company follows the following rules to assess whether a customer should be insured or not and calculate his/her premium rate and maximum amount for which he/she can be insured.



- a. If a person's health is excellent and the person is between 25 and 35 years of age and lives in a city and is a male, then the premium is Rs. 10 per thousand and his policy amount cannot exceed Rs. 200,000.
- b. If a person satisfies all the above conditions except that the gender is female, then the premium is Rs. 13 per thousand and her policy amount cannot exceed Rs. 100,000.
- c. If a person's health is poor and the person is between 25 and 35 years of age and lives in a village and is a male, then the premium is Rs. 16 per thousand and his policy cannot exceed Rs. 50,000.
- d. In all other cases the person is not insured.

Problem # 4:

A library charges a fine for every book returned late. For first 5 days late, the fine is Rs. 50, for 6-10 days late an additional fine of Rs. 100 per day will be charged, and if the late days are above 10 then the fine is Rs. 200 per day. If a book is returned after 30 days, then Rs. 400 per day will be charged and the student's membership will be cancelled. You have to write an application for a library which will take the following inputs and calculate the fine for a student. Inputs: no. of books to be returned, no. of late days for each book (0 would mean the book is returned timely).

Problem # 5:

A university charges \$3000 per semester for in-state tuition and \$4500 per semester for out-of-state tuition. In addition, room and board is \$2500 per semester for in-state students and \$3500 per semester for out-of-state students. Write an application that asks the user for their residential status (i.e., in-state or out-of-state) and whether they require room and board (Y or N). The program should then compute and output their bill for that semester. Use the sample outputs below for a better understanding (do not hard code, your program should run for any input).

Sample Run 1:

Please input "I" if you are in-state or "O" if you are out-of-state:

I

Please input "Y" if you require room and board and "N" if you do not:

N

Your total bill for this semester is \$3000

Sample Run 2:

Please input "I" if you are in-state or "O" if you are out-of-state:

O

Please input "Y" if you require room and board and "N" if you do not:

Y

Your total bill for this semester is \$8000

**Problem # 6:**

You found an exciting summer job for five weeks. It pays, say, \$15.50 per hour. Suppose that the total tax you pay on your summer job income is 14%. After paying the taxes, you spend 10% of your net income to buy new clothes and other accessories for the next school year and 1% to buy school supplies. After buying clothes and school supplies, you use 25% of the remaining money to buy savings bonds. For each dollar you spend to buy savings bonds, your parents spend \$0.50 to buy additional savings bonds for you.

Write a program that prompts the user to enter the pay rate for an hour and the number of hours you worked each week. The program then outputs the following:

- Your income before and after taxes from your summer job.
- The money you spend on clothes and other accessories.
- The money you spend on school supplies.
- The money you spend to buy savings bonds.
- The money your parents spend to buy additional savings bonds for you.

Problem # 7:

You have several pictures of different sizes that you would like to frame. A local picture-framing store offers two types of frames—regular and fancy. The frames are available in white and can be ordered in any color the customer desires. Suppose that each frame is 1 inch wide. The cost of coloring the frame is \$0.10 per inch. The cost of a regular frame is \$0.15 per inch, and the cost of a fancy frame is \$0.25 per inch. The cost of putting a cardboard paper behind the picture is \$0.02 per square inch, and the cost of putting glass on top of the picture is \$0.07 per square inch. The customer can also choose to put crowns on the corners, which costs \$0.35 per crown. Write a program that prompts the user to input the following information and then output the cost of framing the picture:

- The length and width, in inches, of the picture
- The type of the frame
- Customer's choice of color to color the frame
- If the user wants to put the crowns, then the number of crowns

Problem # 8:

Write an application for a bottle vending machine in which a customer inserts an amount and selects the cold drink he/she wants. We assume that the vending machine has five following beverages (7up, Fanta, Sprite, Pepsi, Coke). Show the beverages and their prices (of your choice) to the user. Display the beverage name selected by the user, its cost and remaining amount after purchase of that cold drink. Machine would display an error message: "Out of money" if the amount entered by the user is less than the cost of beverage, he/she wants to purchase.

Problem # 9:

Write an application for your CS1002 instructor who wants to find three students with highest and lowest marks in the mid-term exam. Your application should first ask for the class strength (i.e.,



no. of students), input the mid-exam marks for each student. Find and display the three highest and three lowest marks of the class.

Problem # 10:

The population of a town A is less than the population of town B. However, the population of town A is growing faster than the population of town B. Write an application that prompts the user to enter the population and growth rate of each town. The program outputs after how many years the population of town A will be greater than or equal to the population of town B and the populations of both the towns at that time.

Sample input:

Population of town A = 5000

Growth rate of town A = 4%

Population of town B = 8000

Growth rate of town B = 2%